

ACADEMIC & PATENT ADVISORY DOSSIER

AchiVAI: Patent Evaluation & Provisional Specification Dossier

A Comprehensive Technical Audit, Indian Patent Office (IPO) Form 2 Provisional Specification, and Strategic Intellectual Property Advisory on Deterministic-Semantic ATS Calibration & Closed-Loop Document Synthesis.

PRINCIPAL INVENTOR & AUTHOR

Aman

Lead Developer & System Architect

TARGET INSTITUTION

College / University IPR Cell

Department of Computer Science / Engineering

PROJECT CODEBASE

AchiVAI (Next.js SaaS Platform)

AI-Resume-Enhancer Engine

DOCUMENT VERSION & DATE

v1.0 (Confidential)

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Notice: This document contains proprietary technical formulations, provisional patent specifications under the Indian Patents Act, 1970, and strategic risk analyses. Prepared specifically for consultation with faculty advisors and registered patent attorneys.

EXECUTIVE SUMMARY

1. Executive Briefing for Faculty Advisor

This dossier has been compiled to evaluate whether the **AchiVAI** software project contains patentable intellectual property under Indian Patent Law (The Patents Act, 1970) and international standards, and to present a fully drafted **Provisional Patent Specification (Form 2)** for institutional review.

⚠ The Honest Technical Reality of the Current Code

In its present operational state, AchiVAI is an application wrapper: it extracts text using open-source libraries (`pdf-parse` , `mammoth`), sends it to Google's `gemini-1.5-flash` API with a prompt, and displays the returned JSON score. Under **Section 3(k) of the Indian Patents Act, 1970** and **35 U.S.C. § 101**, "*asking an off-the-shelf AI API to score a resume*" is non-statutory subject matter (an abstract business idea / computer programme per se) and is categorically non-patentable.

✓ The Inventive Elevation Developed in This Dossier

To transform this project into an enforceable, patentable invention, we have architected the **Hybrid Deterministic-Semantic ATS Calibration and Closed-Loop Document Synthesis System**. Instead of trusting an LLM's non-deterministic black-box score, this architecture introduces a **parallel dual-track evaluation pipeline**, a **closed-form mathematical calibration formula**, and a **closed-loop ATS emulator** that tests compiled PDF/UA data streams for 99.9% machine-readability.

Key Decision Points to Discuss with Your Professor

1. Filing a Provisional Patent (Indian Patent Office):

Does our university/college have an active IPR Cell with student patent sponsorship? Filing a provisional application (Form 1 + Form 2) secures an immediate **12-month priority date** at very low statutory cost (₹1,600 for individuals/students), during which we can build out the mathematical calibration benchmarks before filing the Complete Specification.

2. Alternative: High-Impact Academic Conference Publication:

If patent prosecution costs or Section 3(k) hurdles are prohibitive, the exact hybrid architecture formulated in this dossier provides the ideal methodology for an **IEEE / ACM / Springer research paper** titled "*Bridging the Reliability Gap in Automated Resume Screening: A Deterministic-Semantic Calibration Architecture with Closed-Loop Parser Emulation.*"

3. Software Copyright Protection:

Immediate source code copyright registration (Form XIV, Copyright Act, 1957) can be secured for the current Next.js application codebase, protecting the UI, schema, and business logic from plagiarism.

PART 1 • TECHNICAL AUDIT

2. Technical Inventory & Patentability Matrix

Every component of the existing `AchiVAI-AI-Saas_RESUME-ENHANCER` repository was mechanically audited to determine whether novel technical steps exist in the source code.

COMPONENT & SOURCE FILE	MECHANICAL OPERATION	SECTION 3(K) & PRIOR ART RATING	VERDICT
Prompt Construction src/lib/prompt.js	String template concatenation injecting resume text and job description into a prompt instructing Gemini to return JSON with an integer atsScore . No mathematical weighting or chunking.	Novelty: 0/10 Standard prompt engineering. Thousands of open-source wrappers use identical formats. Fails Section 3(k) (mental act / algorithm).	Non-Patentable
Document Text Parser src/lib/fileParsers.js	Invokes npm packages pdf-parse (PDF) and mammoth (DOCX). Applies standard regex to collapse whitespace and remove page tags. Truncates at 30k chars.	Novelty: 0/10 Textbook file I/O using public third-party npm packages. No custom OCR, coordinate trees, or table decoders.	Non-Patentable
End-to-End ATS Scoring src/app/api/analyze-resume/	Authenticates user via Supabase, checks monthly quota in usage_events , calls Gemini 1.5 Flash, parses JSON, and clamps score (0–100).	Novelty: 0/10 Scoring is 100% computed inside third-party LLM weights. No deterministic post-processing or formula.	Non-Patentable
Billing & Quota Guards src/lib/razorpayServer.js	HMAC-SHA256 hash verification (crypto) and standard SQL monthly count queries (created_at >= startOfMonth).	Novelty: 0/10 Standard RFC 2104 cryptographic boilerplate and standard database counter logic.	Non-Patentable
PDF Export Pipeline src/utils/downloadPDF.js	Renders React HTML offscreen, screenshots DOM using html2canvas , normalizes CSS color spaces via 2D canvas probe, slices image at section boundaries, and saves as PNGs in jsPDF .	Novelty: 2.5/10 Slicing DOM screenshots is standard since 2017. Critical Flaw: Produces a raster image PDF with zero text streams, failing non-OCR ATS parsers.	Unviable (Contradicts ATS claim)

The Engineering Paradox in Current Resume Builders

AchiVAI's web interface promises "*ATS-friendly vector PDF export.*" However, mechanical inspection of `src/utls/downloadPDF.js` reveals it uses `html2canvas` to rasterize DOM elements into bitmap canvases, inserting PNG images into `jsPDF`. When an actual corporate ATS (e.g., Taleo, Workday, Greenhouse) ingests this file without optical character recognition (OCR), it extracts **zero characters**. The patent specification drafted herein explicitly rectifies this by claiming a **dual-layer vector synthesis engine with an integrated closed-loop ATS parser emulator**.

PART 2 • INVENTION DISCLOSURE FORM

3. Invention Disclosure Form (IDF)

Document Ref: ACHIVAI-IDF-2026-001 • **Status:** Confidential

3.1 Title of the Invention

System and Method for Deterministic-Semantic Hybrid Evaluation, Calibration, and Machine-Readable Document Synthesis for Automated Applicant Tracking Systems

3.2 Technical Field

Computational linguistics, natural language processing (NLP), document layout graph parsing, and automated recruitment technologies.

3.3 Technical Problems Addressed

- **Probabilistic LLM Non-Determinism & Score Drift:** Generative models yield divergent scores and hallucinated keyword lists across identical resume-JD pairs due to stochastic token sampling.
- **The Client-Side "Image Trap":** Web resume builders screenshot HTML trees into bitmap PDFs, causing total parsing failure in standard enterprise ATS non-OCR pipelines.
- **Absence of an Auditable Calibration Metric:** Enterprise recruitment compliance requires mathematical explainability, which black-box neural networks cannot provide.

3.4 Prior Art Differentiation

- **Jobscan & Lexical Matchers:** Rely solely on exact string matches and n-gram frequencies. They fail to identify contextual semantic equivalence (e.g., "GCP architect" vs "distributed cloud engineering").
- **Generative LLM Wrappers (Resume Worded, Rezi):** Offload scoring entirely to external models via prompts. Unpatentable under Section 3(k) and 35 U.S.C. § 101, non-reproducible, and prone to hallucinations.
- **US Patent 10,242,084 B2:** Discloses rules-based resume entity extraction. It lacks parallel vector projection, closed-form multi-stage calibration, and emulator verification loops.

3.5 System Architecture Overview



4. Form 2: Provisional Patent Specification

FORM 2

THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003

PROVISIONAL SPECIFICATION

(See Section 10 and Rule 13)

4.1 Title of the Invention

A SYSTEM AND METHOD FOR DETERMINISTIC-SEMANTIC HYBRID EVALUATION, CALIBRATION, AND MACHINE-READABLE DOCUMENT SYNTHESIS FOR AUTOMATED APPLICANT TRACKING SYSTEMS

4.2 Preamble to the Description

PROVISIONAL SPECIFICATION: The following specification describes the invention.

4.3 Detailed Description of the Preferred Embodiments

Module 1: Structural Tokenizer & Layout Graph Decomposition

When an electronic candidate file is ingested, the system reads vector character objects, font metrics, and bounding-box coordinates (x, y, w, h) into a spatial proximity cluster tree. Section headers are detected via font weight differentials and mapped to canonical nodes (e.g., `{"EMPLOYMENT HISTORY", "WORK EXPERIENCE"}` → `EXPERIENCE`). Regex automata tag contact endpoints and chronological dates, establishing an auditable document graph.

Module 2: Parallel Dual-Track Evaluation Pipeline

Processing executes across two concurrent, asynchronous channels:

- **Track A (Deterministic Lexical Engine):** Extracts mandatory technical entities (R_1) and preferred entities (R_2) from the job description. Scans the candidate document graph using an exact and lemmatized n-gram sliding window ($n \in \{1, 2, 3, 4\}$) with bi-directional acronym expansion tables (e.g., `"Kubernetes" ↔ "K8s"`). Lexical coverage is computed as:

$$S_{\text{lex}} = 100 \times \left(\lambda_1 \frac{|T_{\text{res}} \cap R_1|}{|R_1|} + \lambda_2 \frac{|T_{\text{res}} \cap R_2|}{|R_2|} \right)$$

- **Track B (Dense Vector Semantic Projection):** Projects experience statements v_i and requirement clauses u_j into dense vector embeddings. Computes a pairwise cosine similarity matrix $C_{ij} = \frac{v_i \cdot u_j}{\|v_i\| \|u_j\|}$. The mean of the top- k aligned requirements determines the semantic alignment score S_{sem} .

Module 3: Closed-Form Deterministic Calibration Engine

To prevent LLM hallucination and score drift, the ATS score is sequestered within an explicit closed-form calibration function:

$$S_{\text{ATS}} = \text{clamp}(\alpha \cdot S_{\text{lex}} + \beta \cdot S_{\text{sem}} + \gamma \cdot S_{\text{struct}} - \sum_{k=1}^m \delta_k \cdot P_k, 0, 100)$$

Where S_{struct} represents structural hierarchy compliance, P_k denotes specific penalties (e.g., missing metrics, high passive-verb ratios, omitted Tier-1 skills), and α, β, γ are normalized weights ($\alpha + \beta + \gamma = 1.0$). The resulting score is **100% reproducible across distributed computing instances**.

Module 4: Constrained Generative Residual Synthesizer

The calibration engine outputs a discrete **Residual Vector** $\mathbf{R} = \{K_{\text{missing}}, B_{\text{low-metric}}\}$. The generative language model is restricted strictly to proposing linguistic revisions for the specific identified deficits. The LLM is prohibited from modifying the calculated score or generating unverified factual claims.

Module 5: Closed-Loop ATS-Safe Document Synthesis Engine

The synthesis engine serializes structured profile data into a Portable Document Format (PDF) containing a linearized Unicode text stream with embedded `ToUnicode` CMap tables (PDF/UA ISO standard). Prior to user release, the generated file is processed by an internal **ATS Parser Emulator** (hosting Apache Tika / Poppler extraction harnesses). A fidelity index is computed:

$$\text{Fidelity}(T_{\text{source}}, T_{\text{emulated}}) = \frac{|T_{\text{source}} \cap T_{\text{emulated}}|}{|T_{\text{source}} \cup T_{\text{emulated}}|}$$

If $\text{Fidelity} < 0.999$, the engine automatically reconfigures layout bounding boxes and font tables until complete machine-readability is verified.

4.4 Draft Claims (For Scope & Priority Establishment)

1. (Independent Method Claim)

A computer-implemented method for deterministically evaluating and calibrating applicant profile compatibility against a target job description, comprising: ingesting an electronic candidate document and target job description; decomposing said document into a hierarchical layout graph; executing a dual-track evaluation pipeline comprising (a) a deterministic lexical engine computing an exact n-gram coverage score (S_{lex}) with acronym expansion, and (b) a vector semantic engine computing a contextual cosine alignment score (S_{sem}); calculating a calibrated ATS compatibility score (S_{ATS}) within a closed-form calibration engine combining said lexical score, said semantic score, a structural score (S_{struct}), and parameterized penalties; and generating an explainable output comprising said score and a discrete residual vector of verified lexical omissions.

2. (Dependent Claim • Emulator Loop)

The method as claimed in claim 1, further comprising: synthesizing a candidate resume document into a vector PDF comprising a linearized Unicode text stream; processing said synthesized document through an integrated ATS parser emulator to extract an emulated text stream; computing an extraction fidelity metric against source profile data; and conditionally reconfiguring document layout parameters when said fidelity fails to satisfy a predetermined acceptance threshold.

3. (Dependent Claim • Calibration Formula)

The method as claimed in claim 1, wherein said closed-form calibration engine computes $S_{\text{ATS}} = \text{clamp}(\alpha S_{\text{lex}} + \beta S_{\text{sem}} + \gamma S_{\text{struct}} - \sum_{k=0}^{100} \delta_k P_k, 0, 100)$, wherein $\alpha + \beta + \gamma = 1.0$, and P_k comprises quantified metric deficiency penalties and structural header deficiency penalties.

4. (Dependent Claim • Constrained Residual LLM)

The method as claimed in claim 1, wherein said discrete residual vector is provided to a constrained generative language model to produce targeted linguistic revisions of candidate achievement statements, wherein said generative language model is restricted from altering said calibrated compatibility score.

5. (Independent System Claim)

A computerized applicant tracking evaluation system, comprising one or more processors and non-transitory computer-readable memory having instructions configured to execute the method of claims 1 to 4.

5. Legal Risks & Strategic Roadmaps

5.1 The Four Primary Risks to Discuss with Patent Counsel

- **1. Section 3(k) Objection (Indian Patents Act):**
The Indian Patent Office routinely objects to software claims under Section 3(k) as "*computer programmes per se*" or "*business methods*." Defense requires demonstrating a **"technical effect"**: specifically, the mathematical elimination of neural network stochastic variance and the physical closed-loop testing of PDF/UA byte streams by an internal emulator.
- **2. Prior Art from HR-Tech Corporations:**
Companies like Workday, Oracle/Taleo, and Jobscan hold patents on taxonomy parsing and keyword frequency scoring. Our patent relies on the **novel hybrid reconciliation** of deterministic lexical tokens, dense embeddings, and the closed-loop compilation emulator.
- **3. The 12-Month "Provisional-to-Complete" Clock:**
Under Indian patent law, filing a provisional gives you exactly **12 months** to file the Complete Specification. During this year, the mathematical calibration engine and emulator must be built and empirically benchmarked.
- **4. Resolving the "Image-PDF" Flaw in Current Code:**
The existing `downloadPDF.js` relies on `html2canvas` screenshotting. To match the patent claims, this must be upgraded to a true vector engine (e.g., `@react-pdf/renderer`) that embeds native Unicode text streams.

5.2 Strategic Comparison: Three Alternate Paths

DIMENSION	OPTION A: PROVISIONAL PATENT	OPTION B: RESEARCH PAPER	OPTION C: SOFTWARE COPYRIGHT
Core Goal	Commercial monopoly & IP priority date	Academic prestige & citations	Protection against code theft
Legal Barrier	High (Section 3(k) CRI examination)	Peer review (methodology & data)	Minimal (formal registration)
Official Fees	₹1,600 (Student) / ₹8,000 (Entity)	Zero to standard conference fee	₹500 (Individual)
Ideal For	College IPR sponsorship / Startup IP	M.Tech / B.Tech / PhD thesis credit	Immediate SaaS code protection

5.3 Recommended Next Steps for Aman

1. **Print or share this complete PDF dossier** with your project guide or faculty head.
2. **Visit your University IPR Cell:** Request an evaluation for filing Form 1 and Form 2 under student sponsorship.
3. **Implement the Baseline Calibration Code:** Introduce the basic mathematical formula in `src/app/api/analyze-resume/route.js` to possess working prototype data.
4. **Switch PDF Export:** Replace `html2canvas` with structured vector rendering to ensure true machine-readability.

End of Patent Dossier • Prepared by Aman • AchiVAI Platform Architecture • 2026